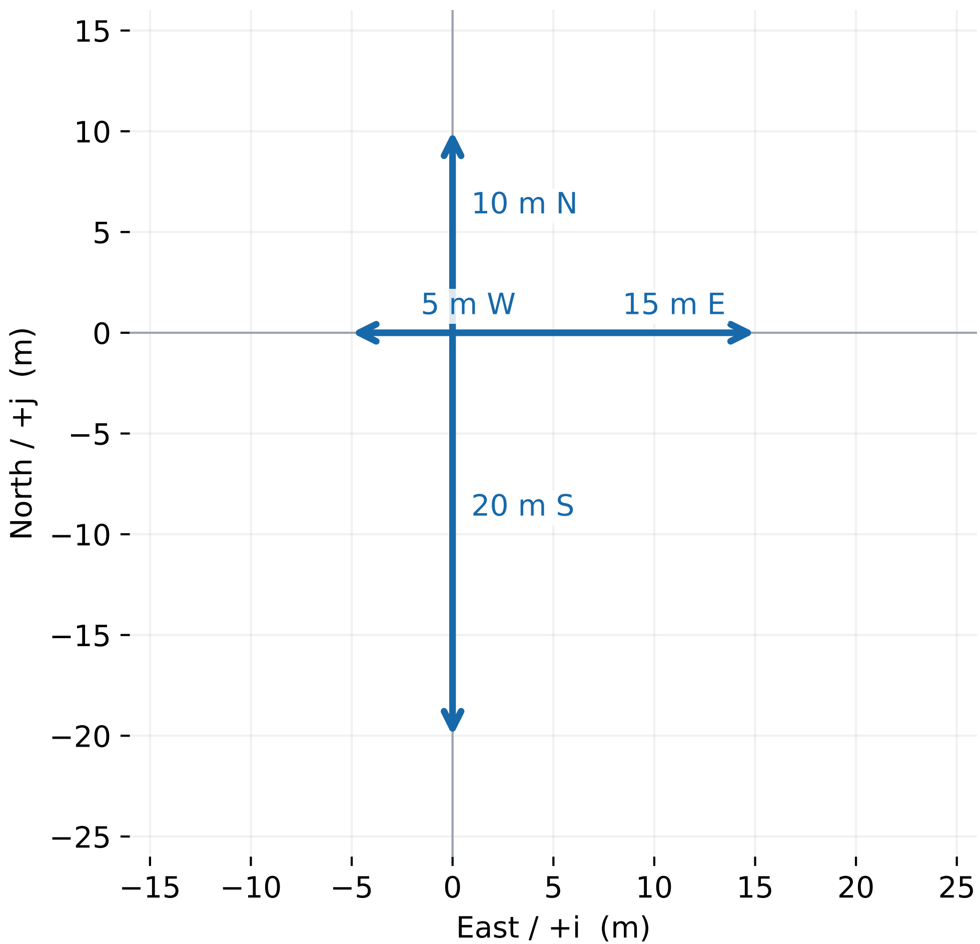
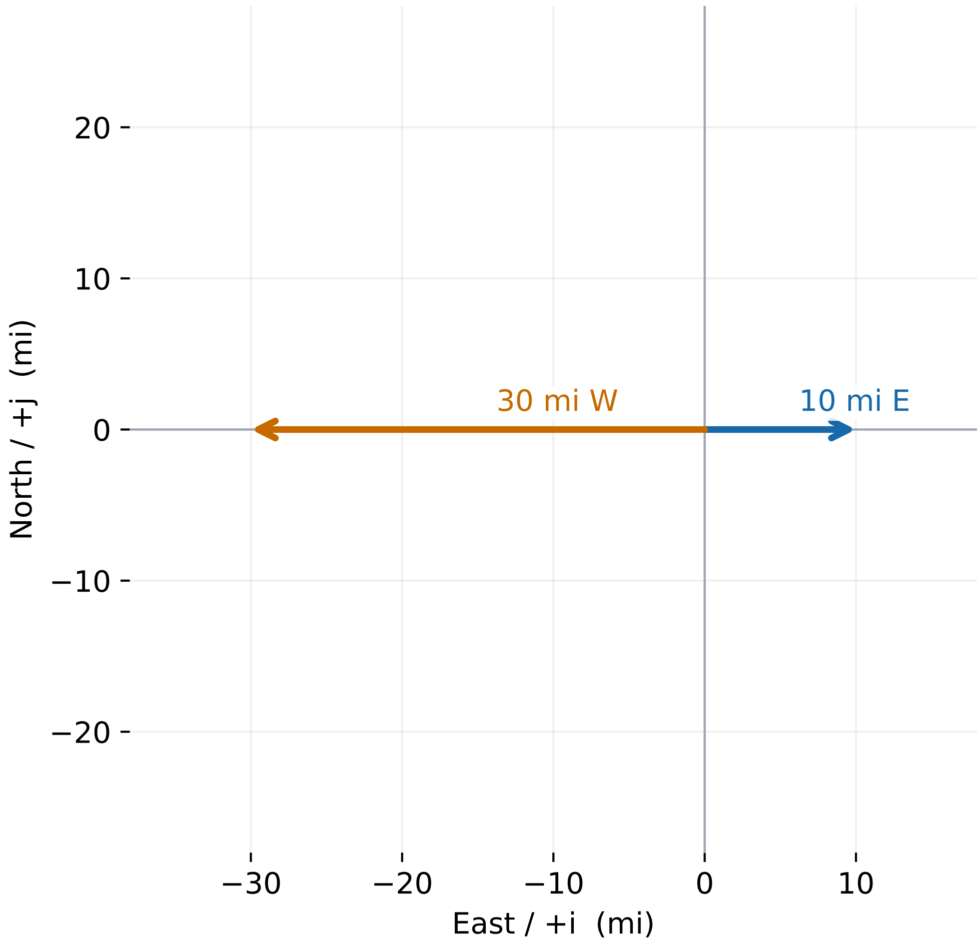
Drawing vectors: introductory example



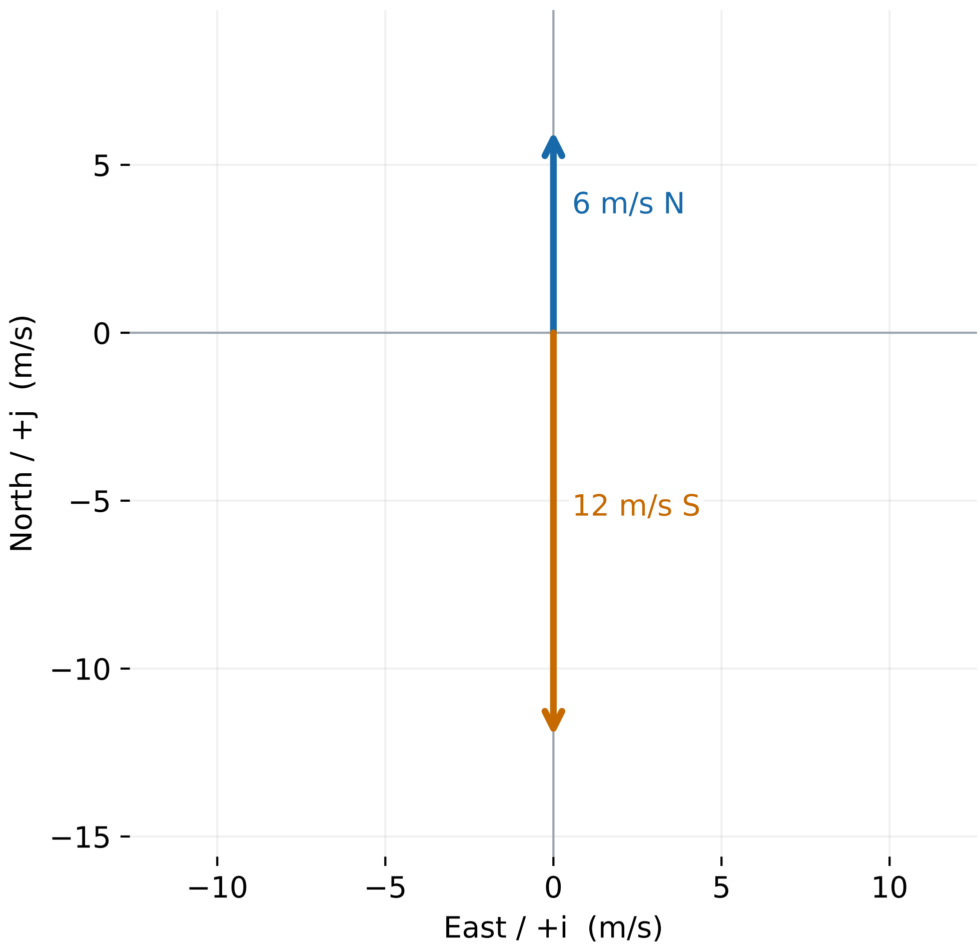
Equal horizontal and vertical scales. North is up; East is right.

Drawing vectors — problem 1



Equal horizontal and vertical scales. North is up; East is right.

Drawing vectors — problem 2



Equal horizontal and vertical scales. North is up; East is right.

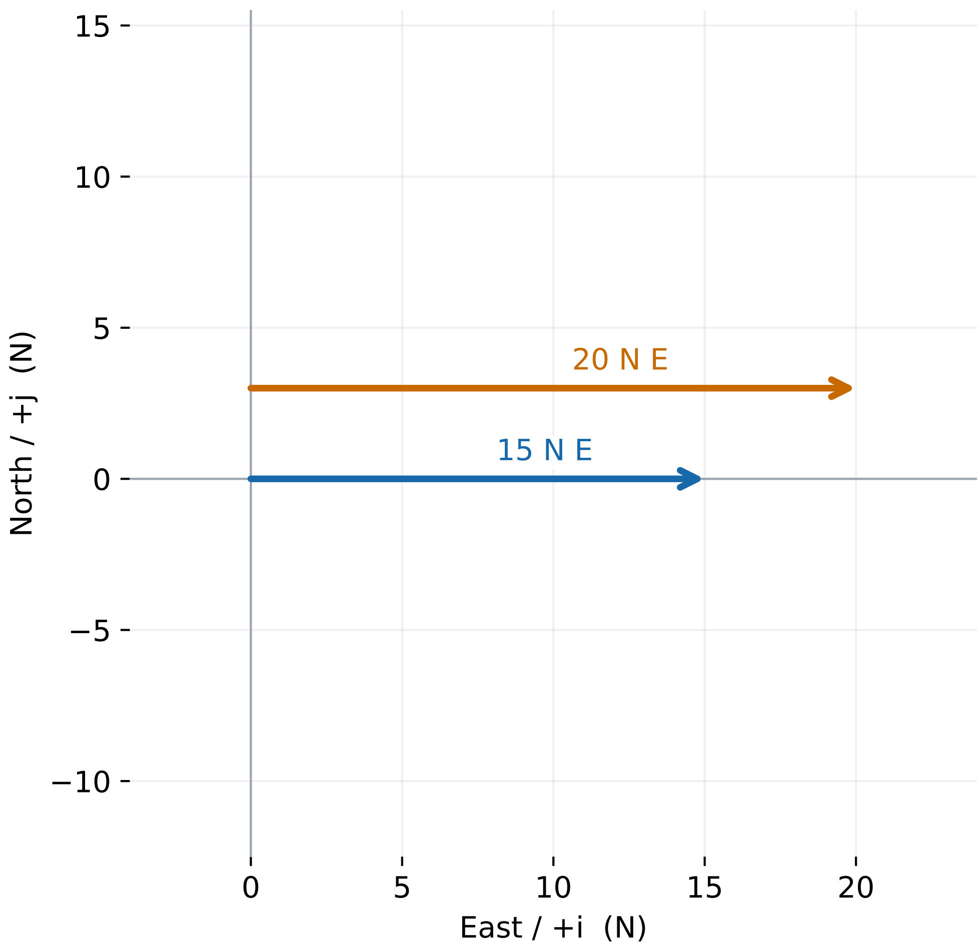
Drawing vectors — problem 3

4 gallons and 20 gallons

Volume is scalar.
No vector arrows can be drawn.

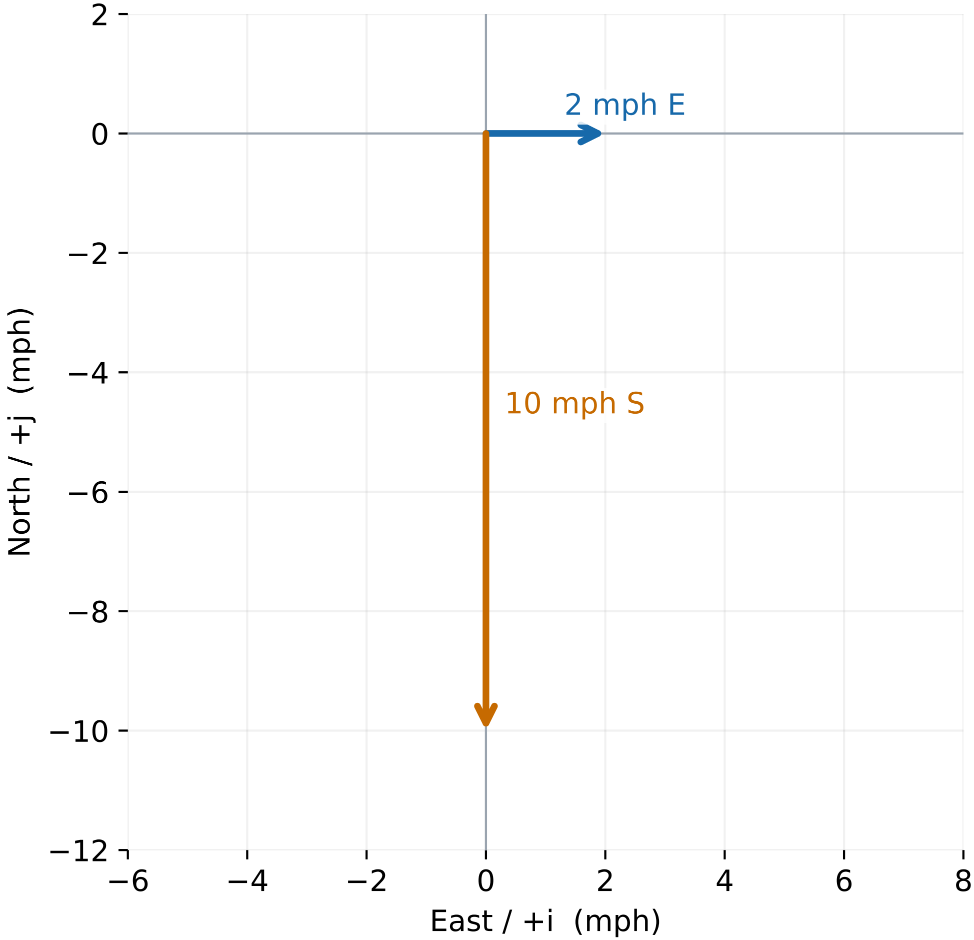
Equal horizontal and vertical scales. North is up; East is right.

Drawing vectors — problem 4



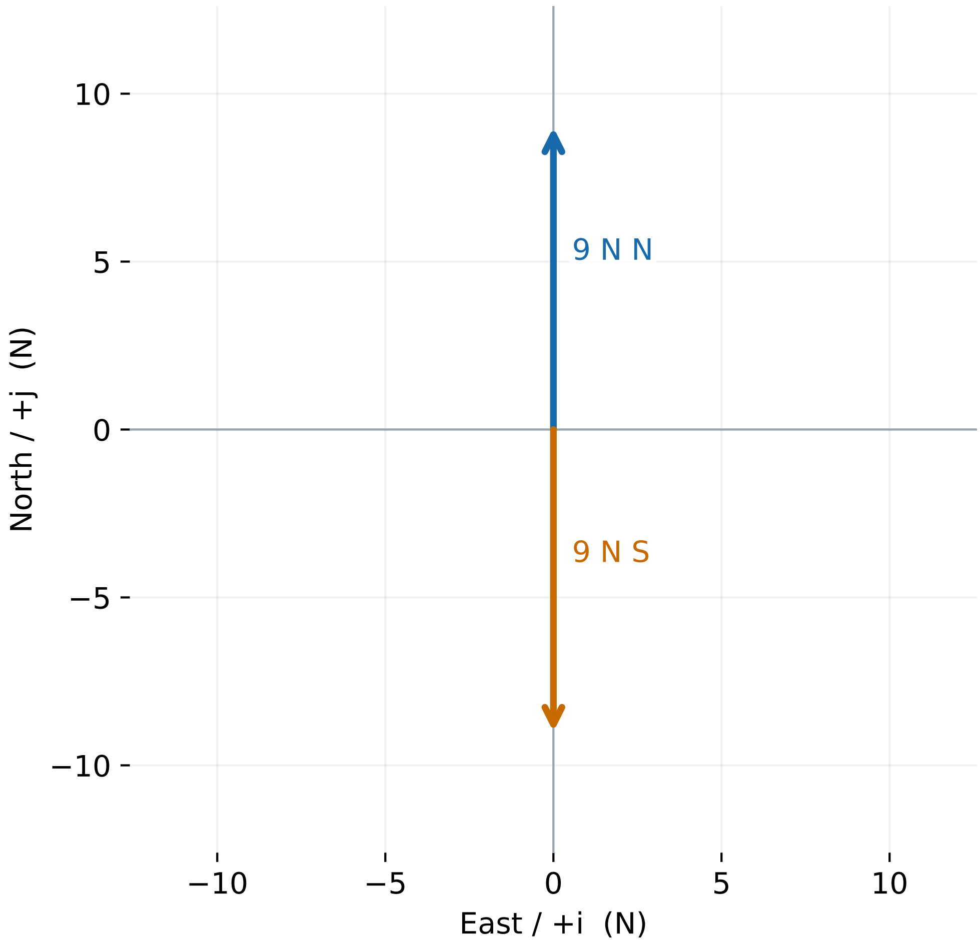
Parallel arrows are offset for visibility; lengths use the same scale.

Drawing vectors — problem 5



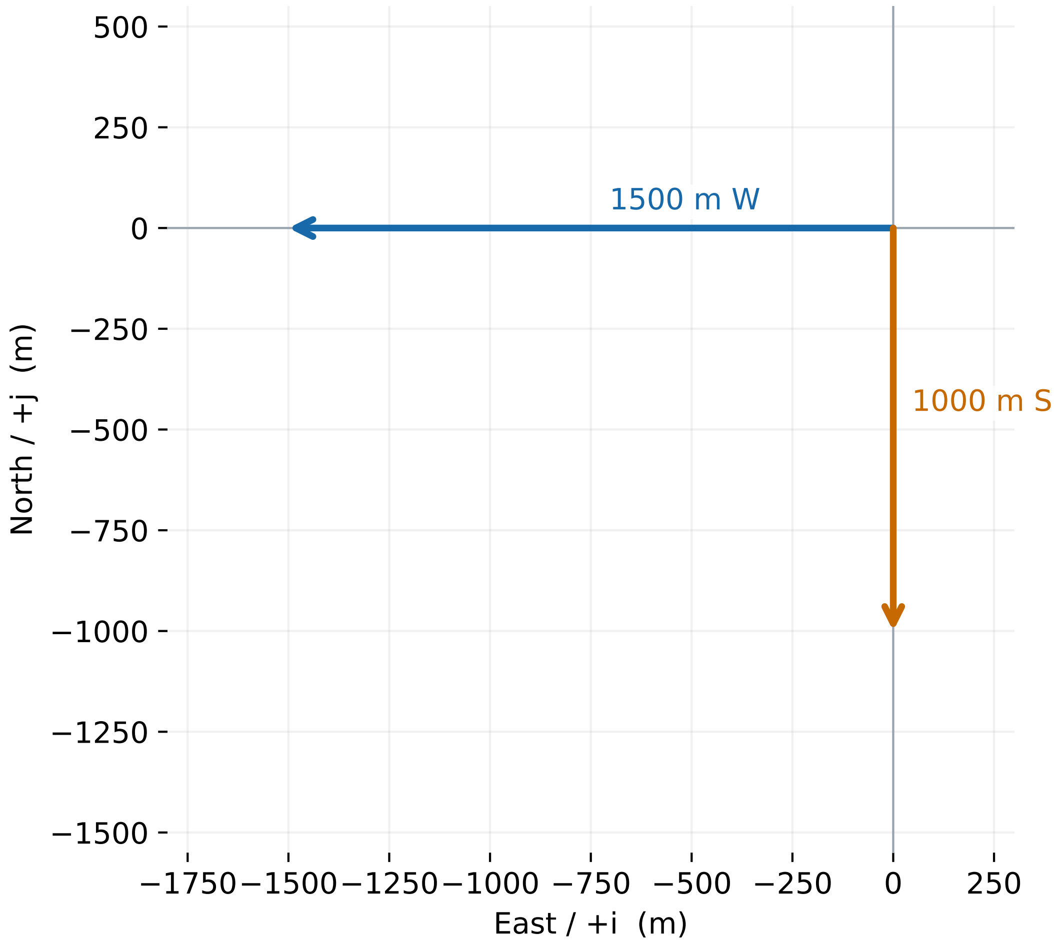
Equal horizontal and vertical scales. North is up; East is right.

Drawing vectors — problem 6



Equal horizontal and vertical scales. North is up; East is right.

Drawing vectors — problem 7



Equal horizontal and vertical scales. North is up; East is right.

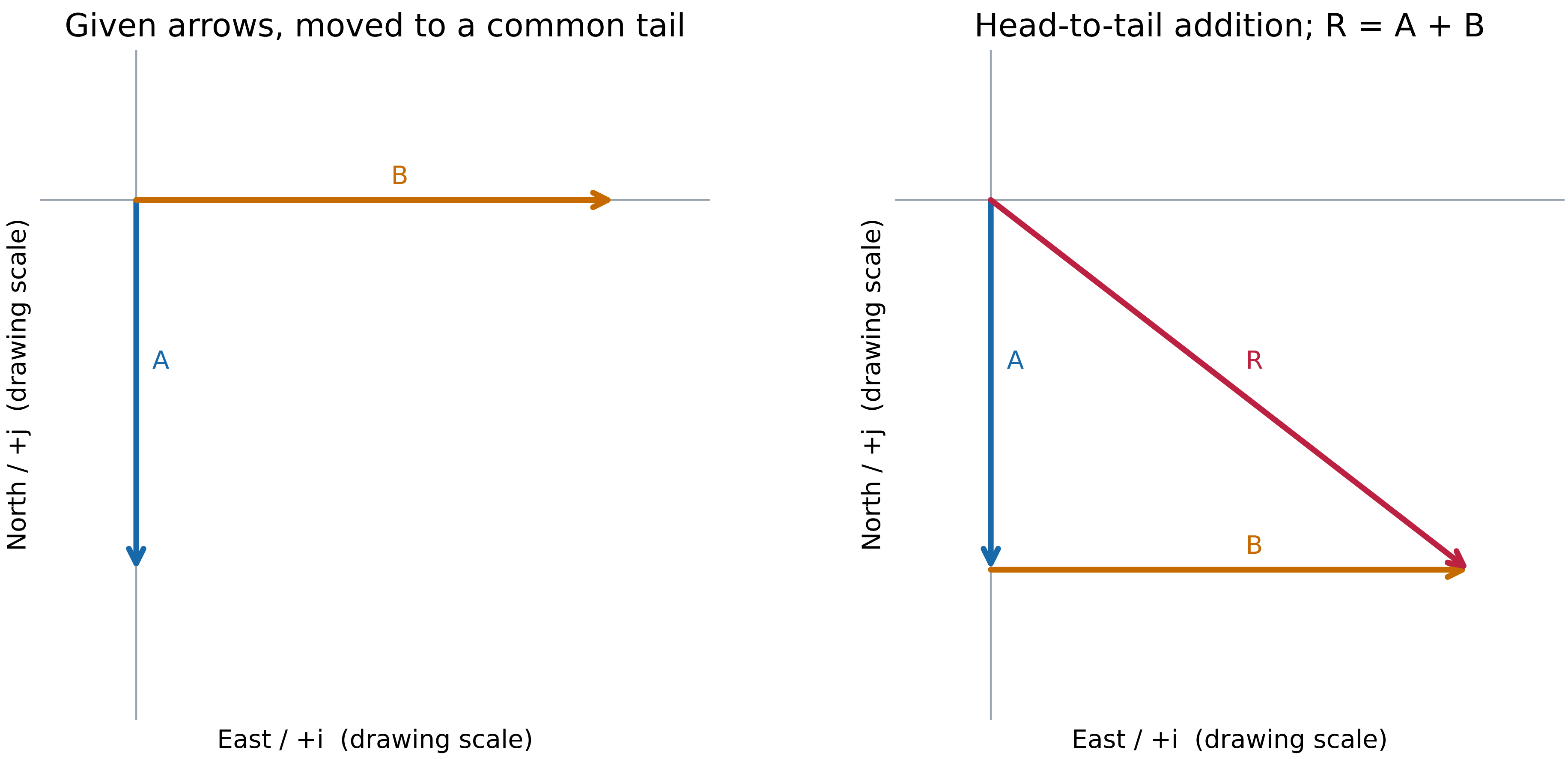
Drawing vectors — problem 8

7 mph and 28 mph

No directions are supplied.
These are speeds, so vector arrows cannot be drawn.

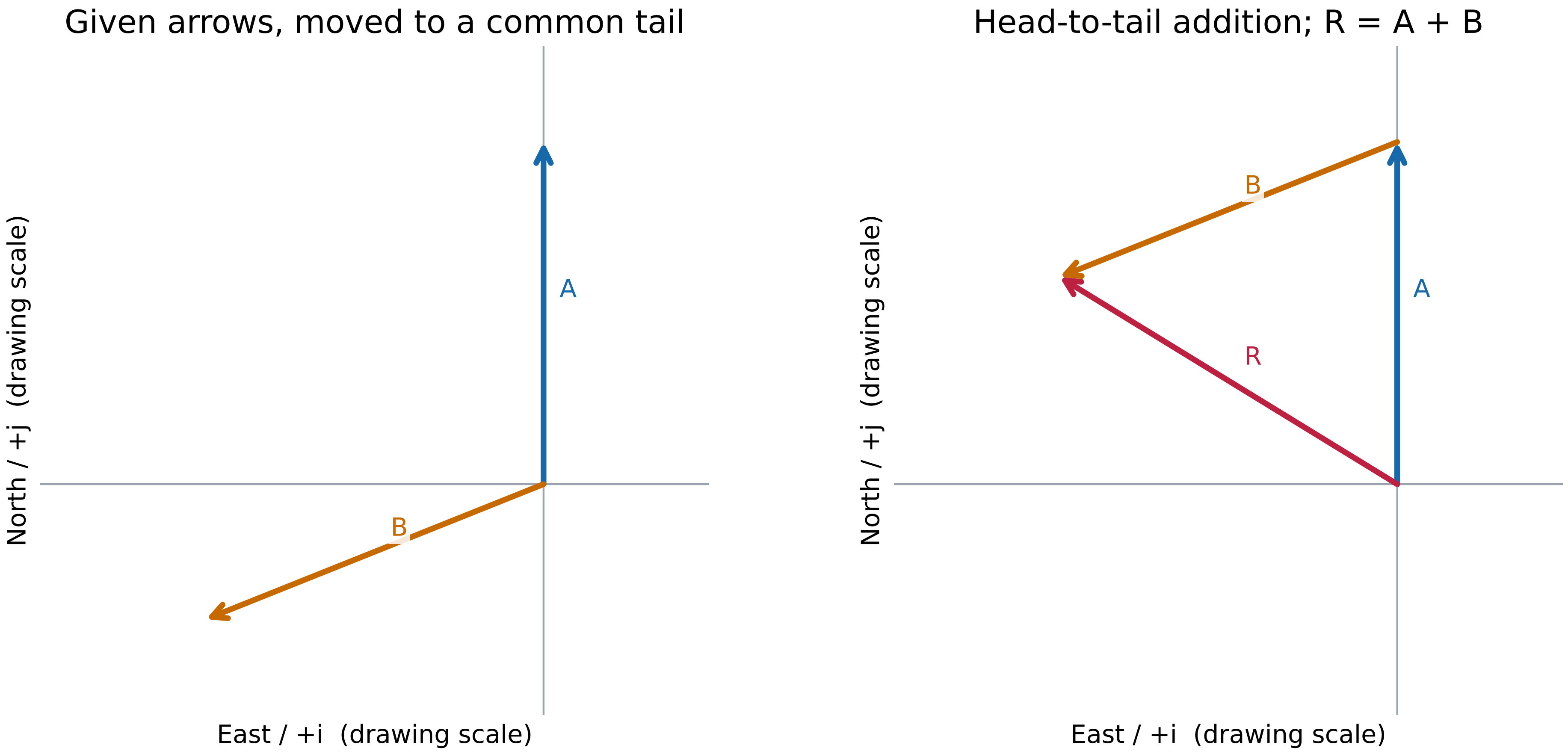
Equal horizontal and vertical scales. North is up; East is right.

Graphical vector addition — problem 1



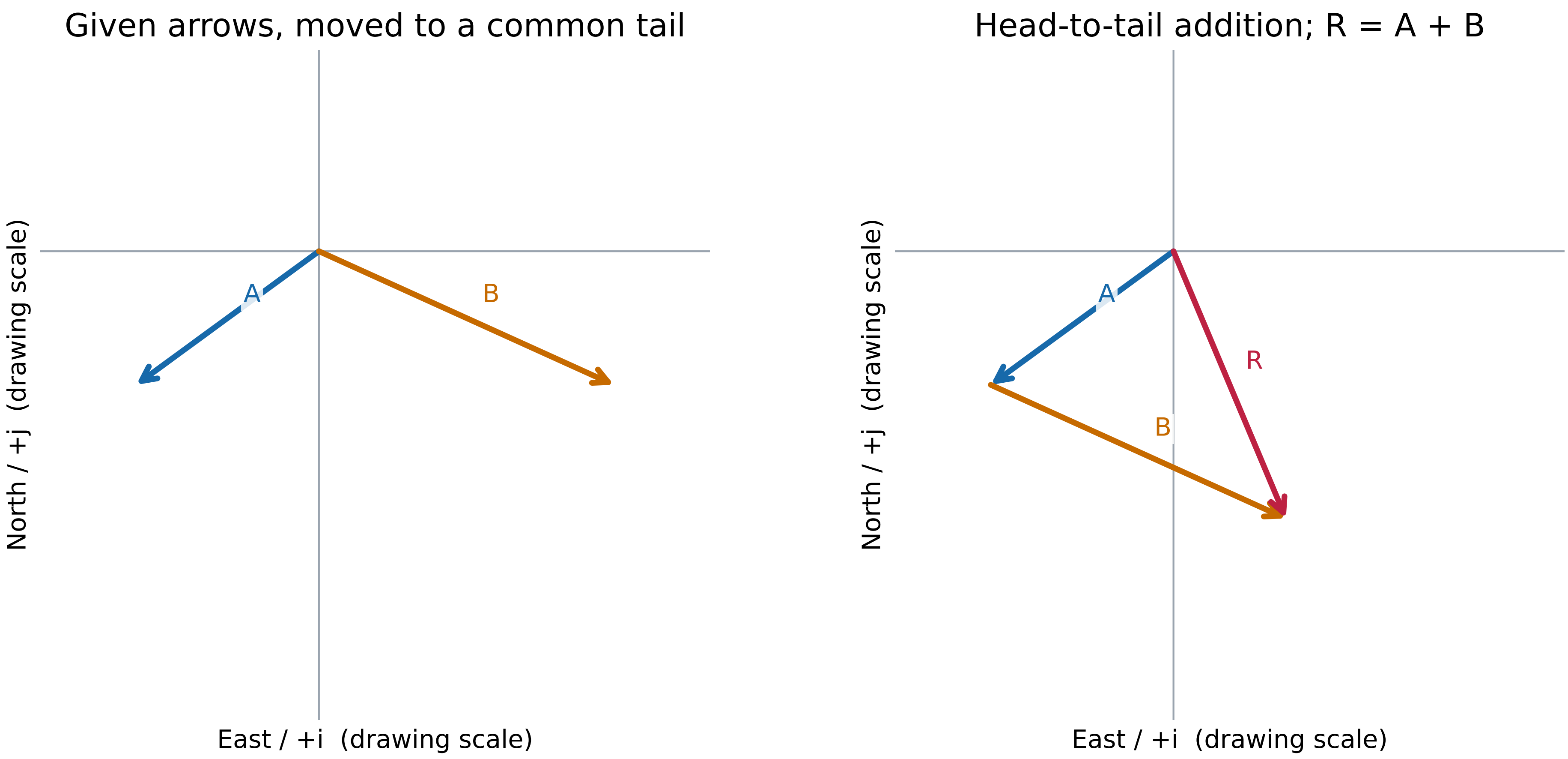
Unlabeled source arrows: approximate scan proportions; no numerical magnitude is given.

Graphical vector addition — problem 2



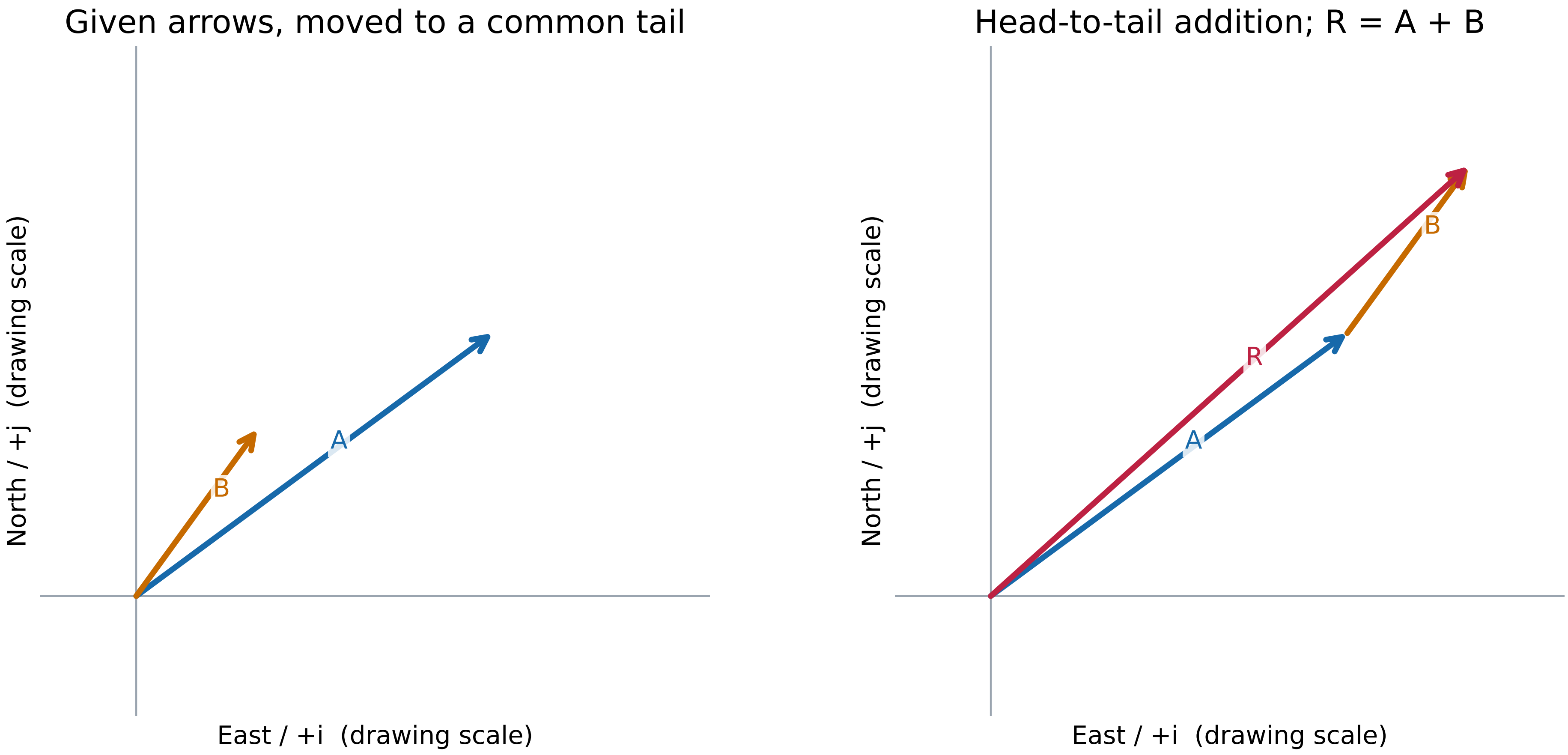
Unlabeled source arrows: approximate scan proportions; no numerical magnitude is given.

Graphical vector addition — problem 3



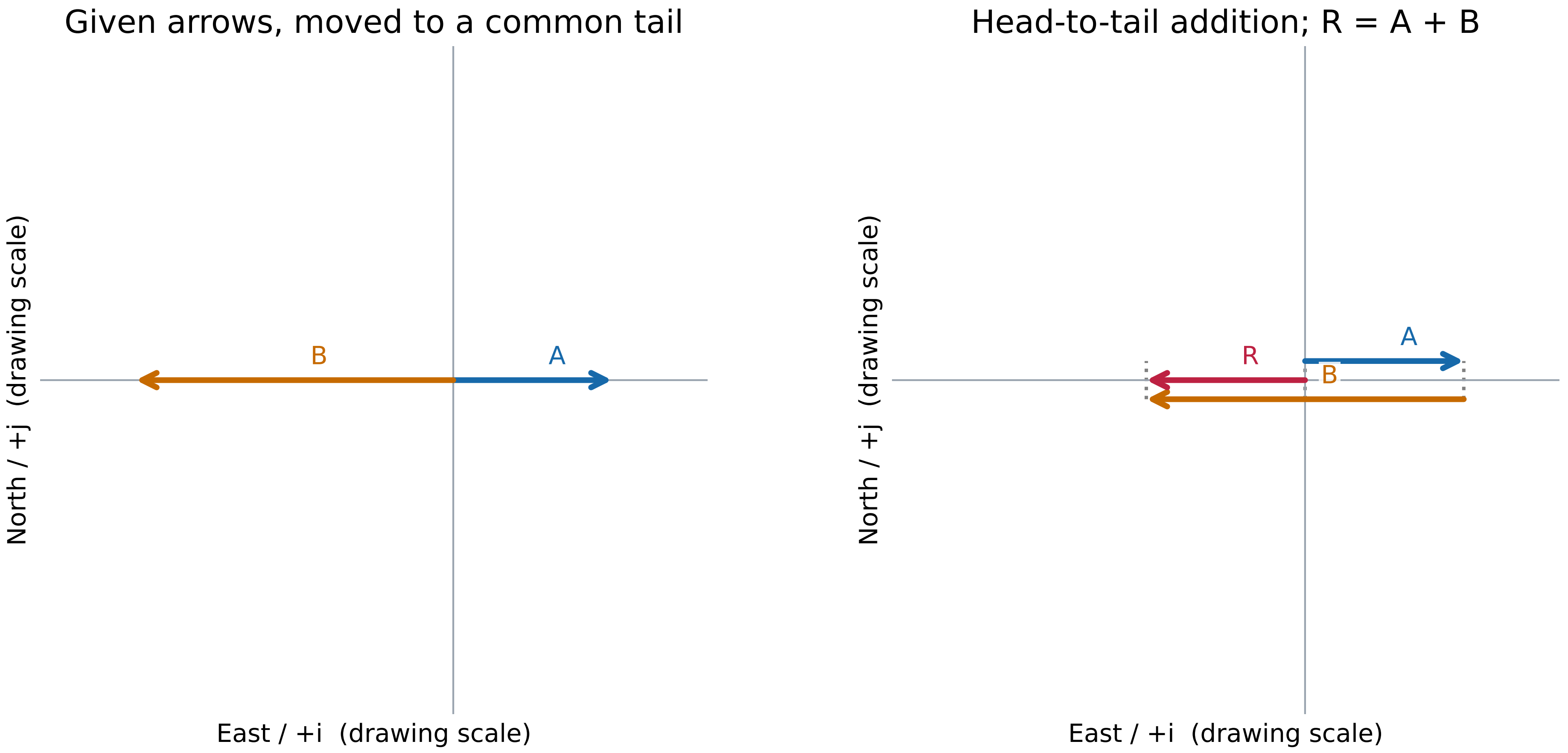
Unlabeled source arrows: approximate scan proportions; no numerical magnitude is given.

Graphical vector addition — problem 4



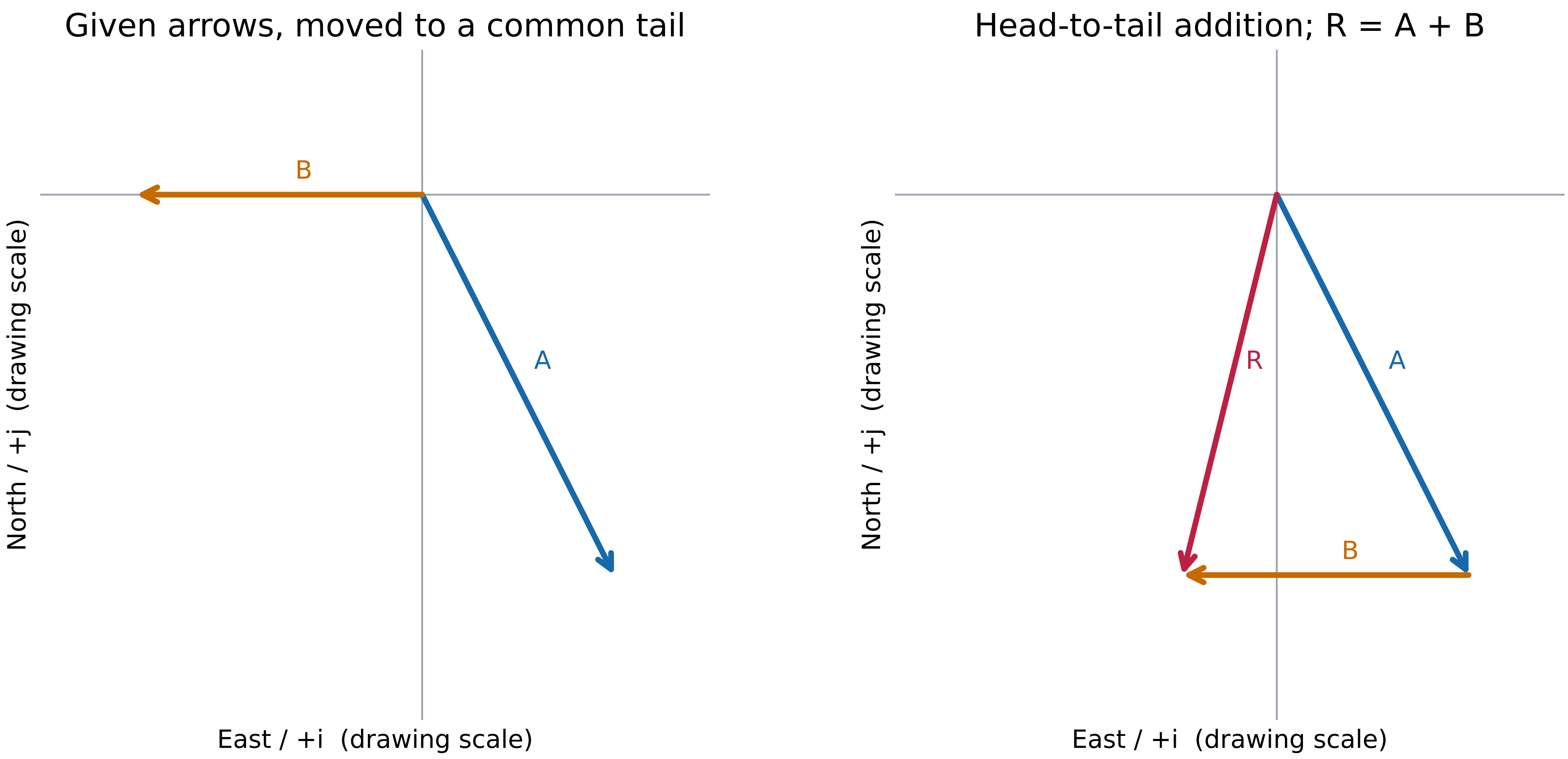
Unlabeled source arrows: approximate scan proportions; no numerical magnitude is given.

Graphical vector addition — problem 5



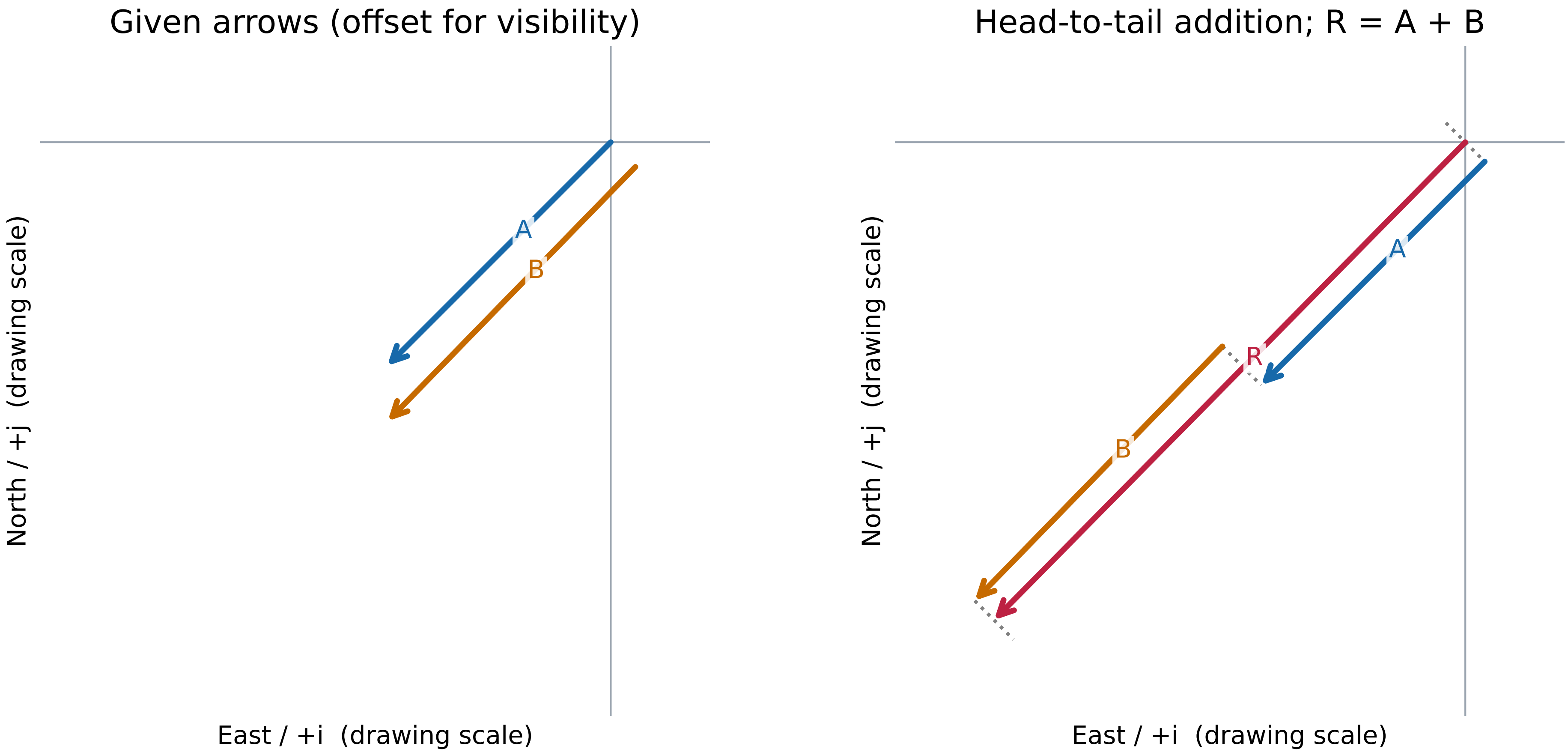
Unlabeled source arrows: approximate scan proportions; no numerical magnitude is given.
Parallel lanes separate overlapping arrows; dotted guides align endpoints.

Graphical vector addition — problem 6



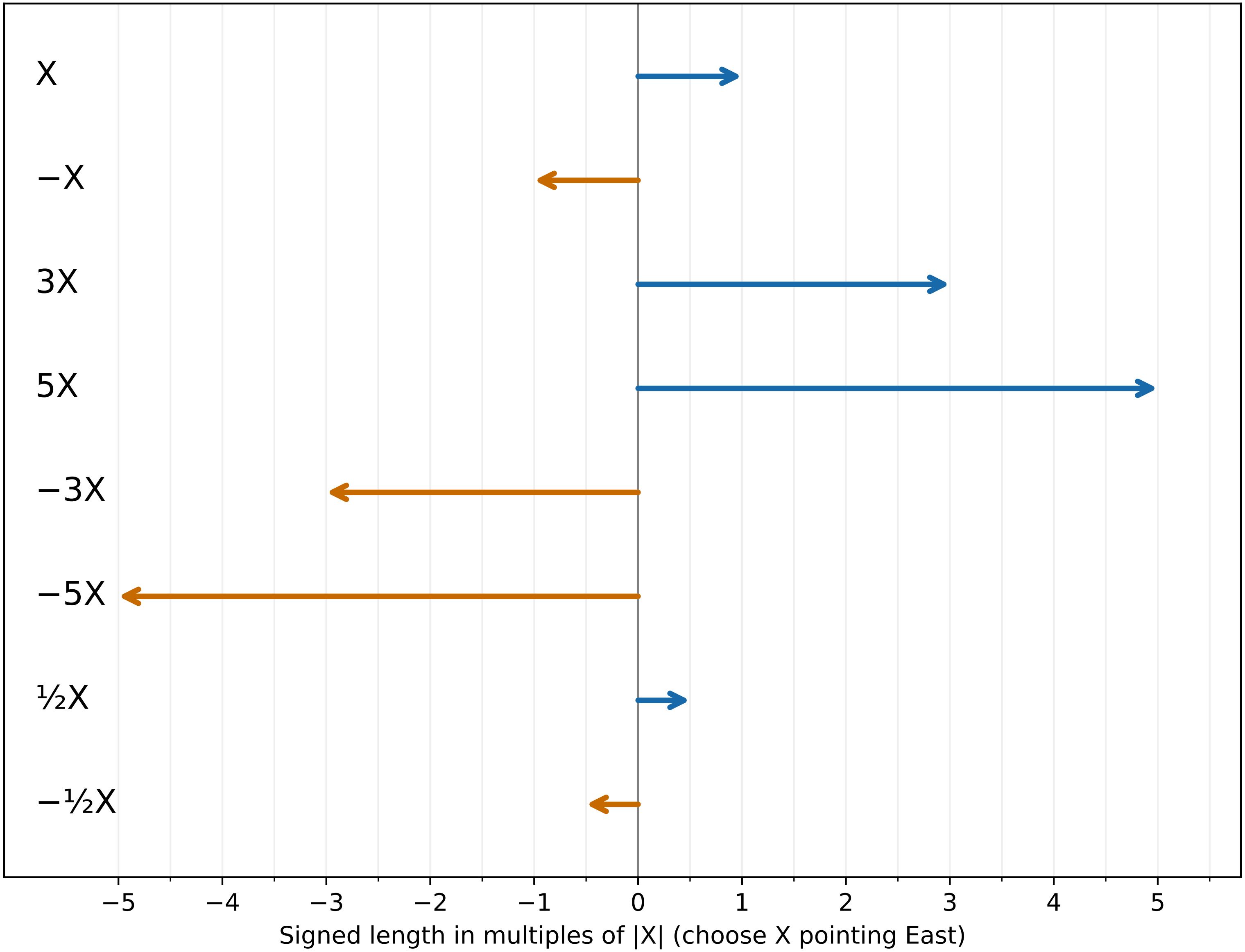
Unlabeled source arrows: approximate scan proportions; no numerical magnitude is given.

Graphical vector addition — problem 7



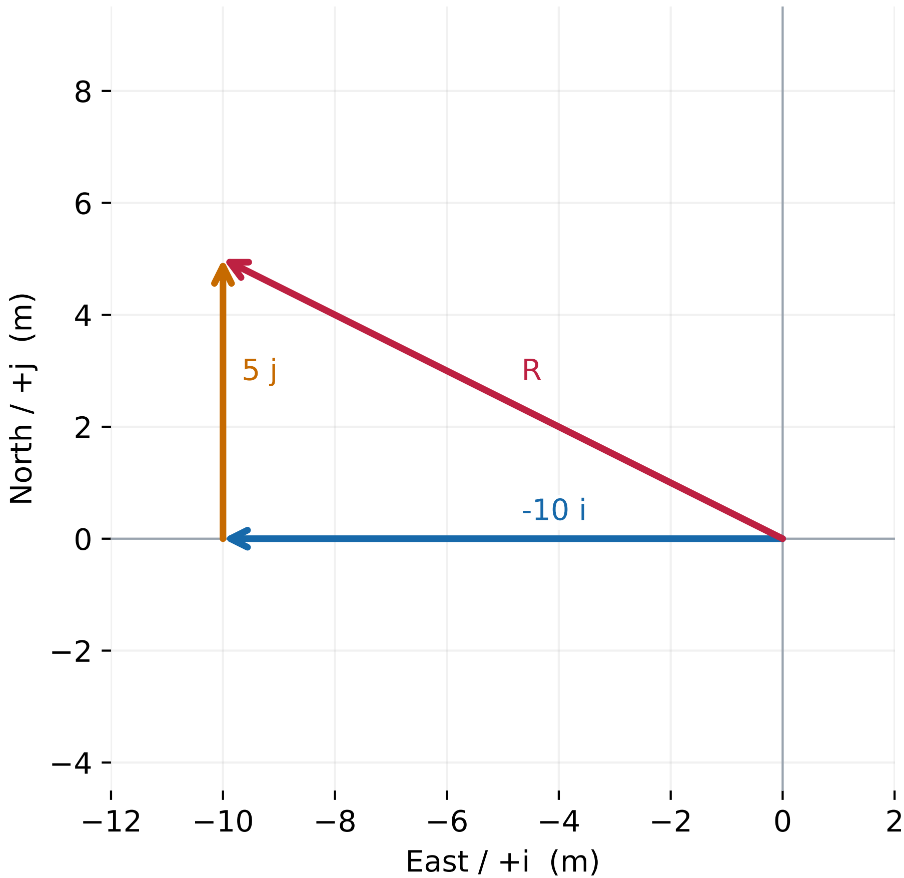
Unlabeled source arrows: approximate scan proportions; no numerical magnitude is given.
Parallel lanes separate overlapping arrows; dotted guides align endpoints.

Scalar multiplication — all eight vectors



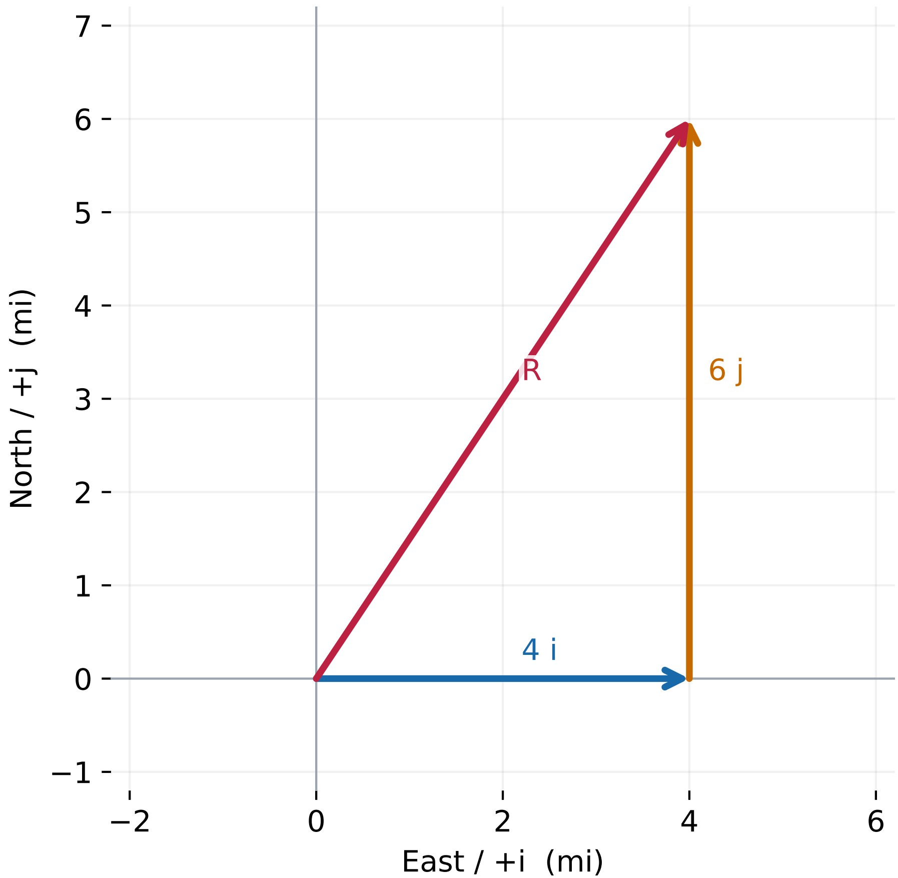
Positive multiples keep direction; negative multiples reverse it. All arrows share one scale.

Vector components — problem 1
R = (-10 i + 5 j) m



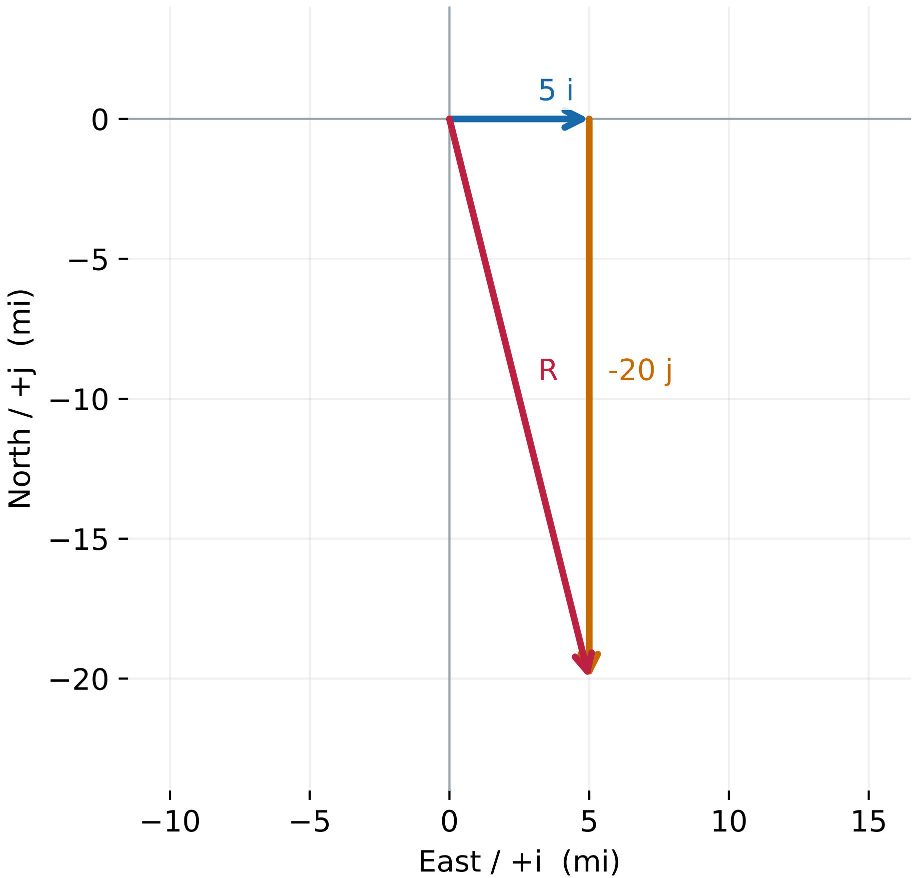
Equal horizontal and vertical scales. North is up; East is right.

Vector components — problem 2
R = (4 i + 6 j) mi



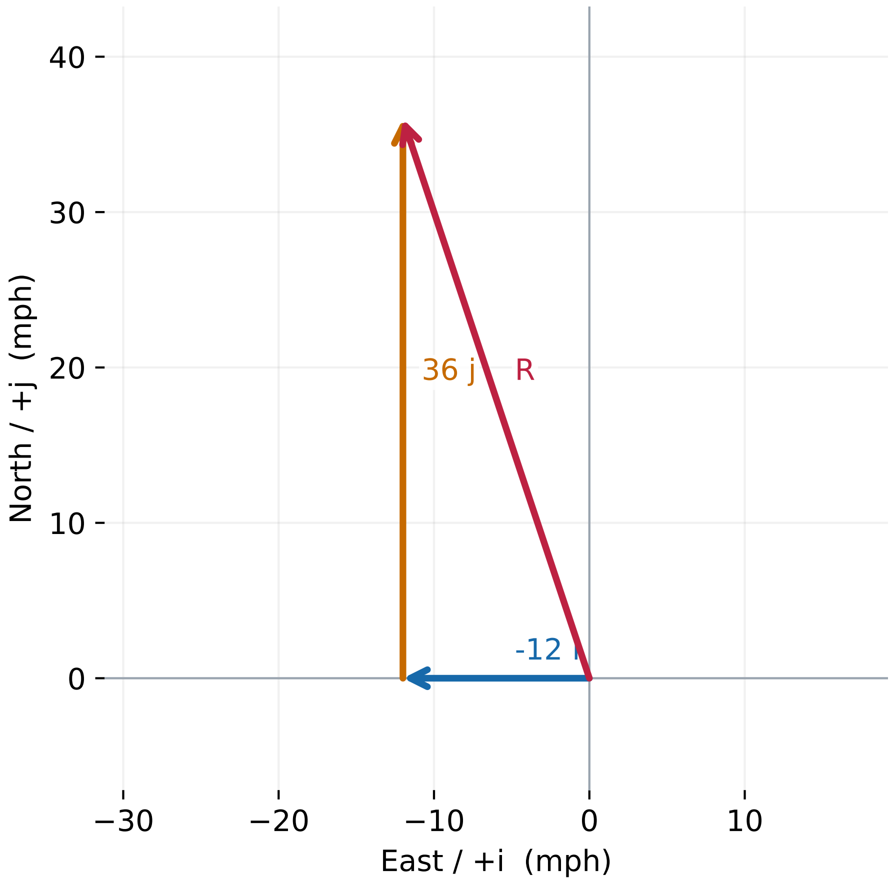
Equal horizontal and vertical scales. North is up; East is right.

Vector components — problem 3
R = (5 i − 20 j) mi



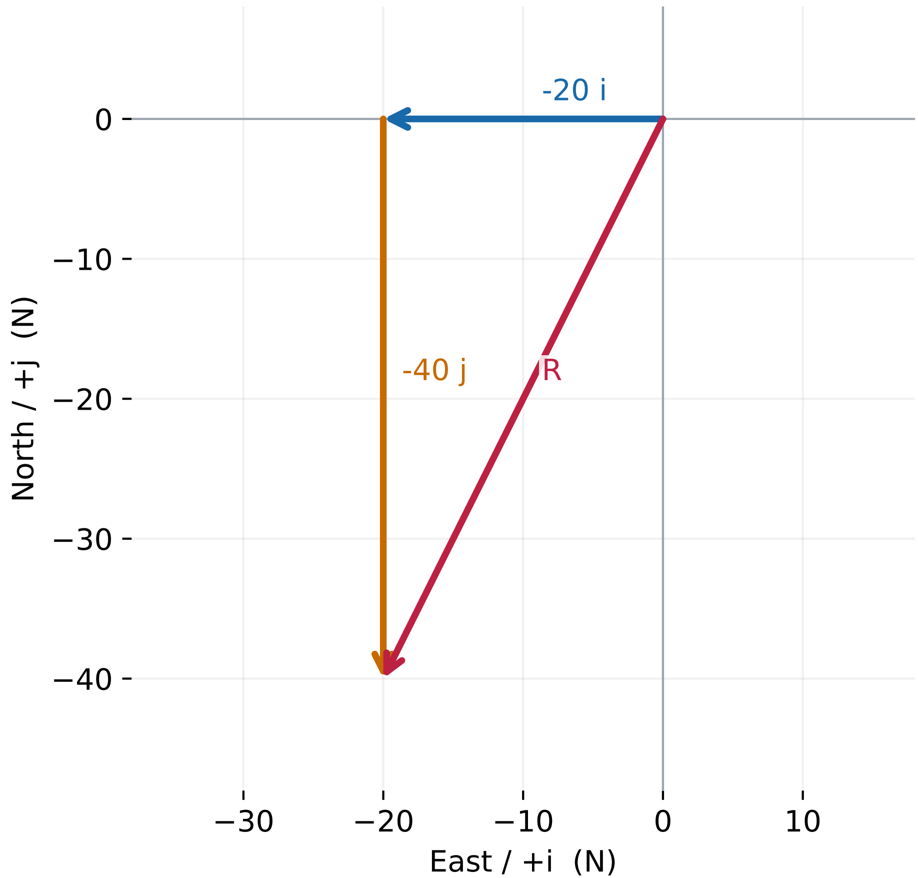
Equal horizontal and vertical scales. North is up; East is right.

Vector components — problem 4
R = (-12 i + 36 j) mph



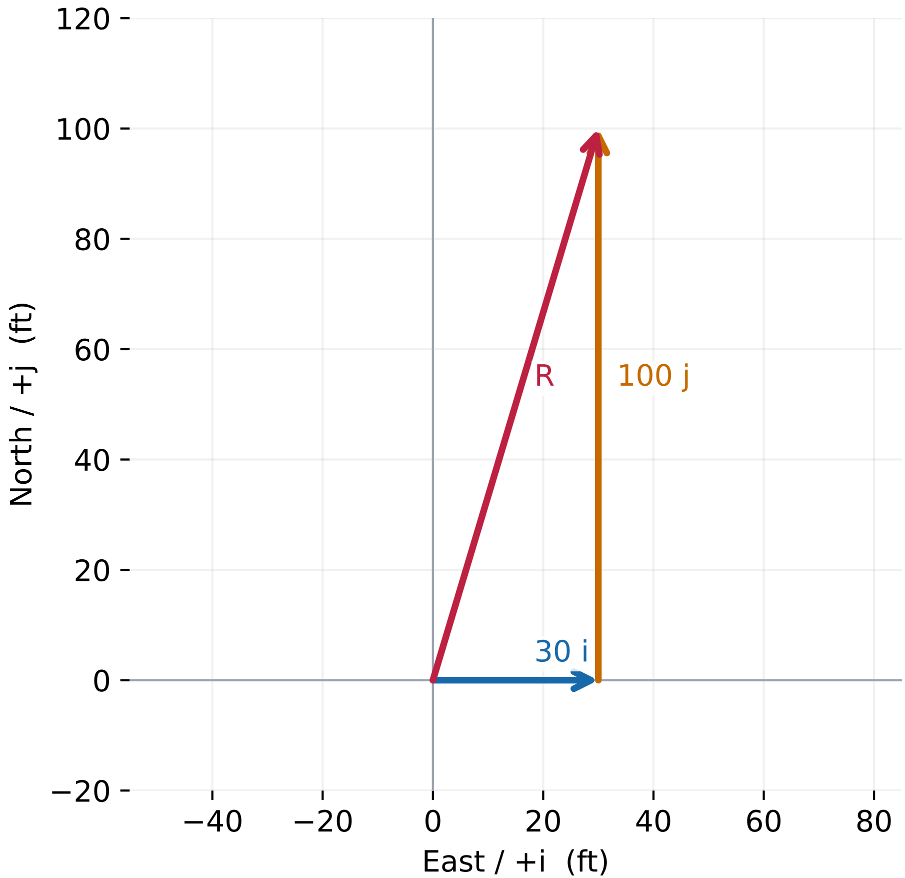
Equal horizontal and vertical scales. North is up; East is right.

Vector components — problem 5
R = (-20 i − 40 j) N



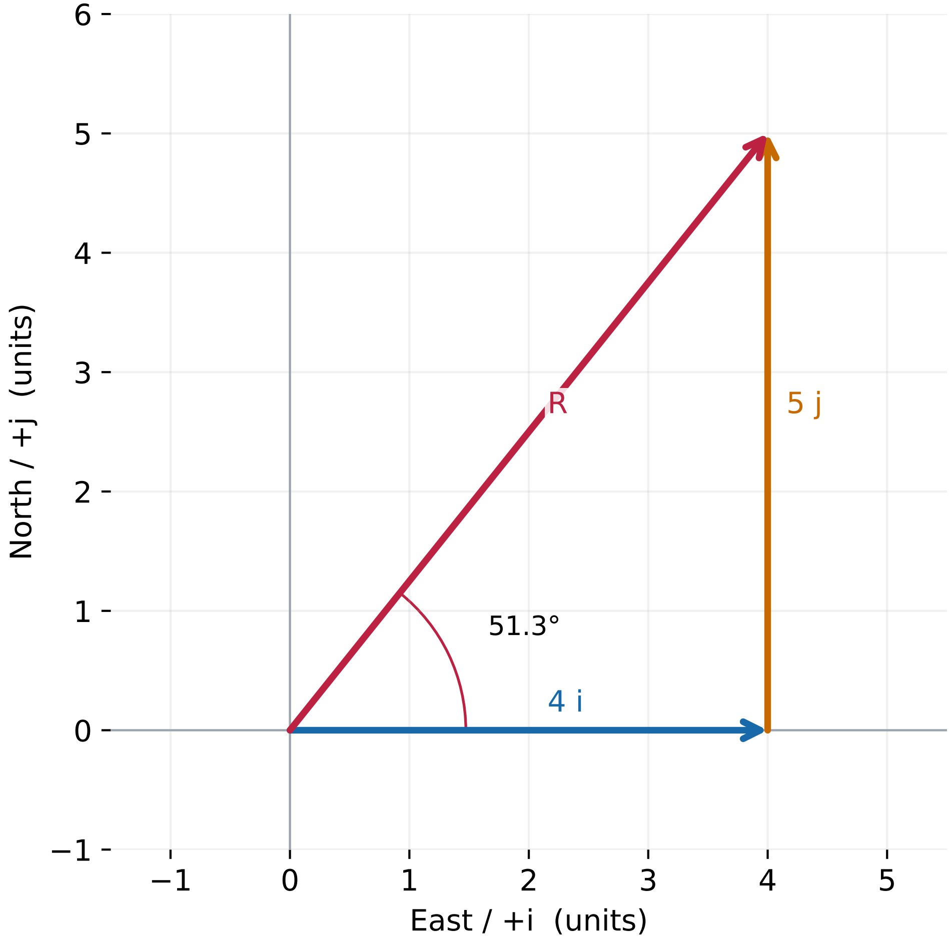
Equal horizontal and vertical scales. North is up; East is right.

Vector components — problem 6
R = (30 i + 100 j) ft



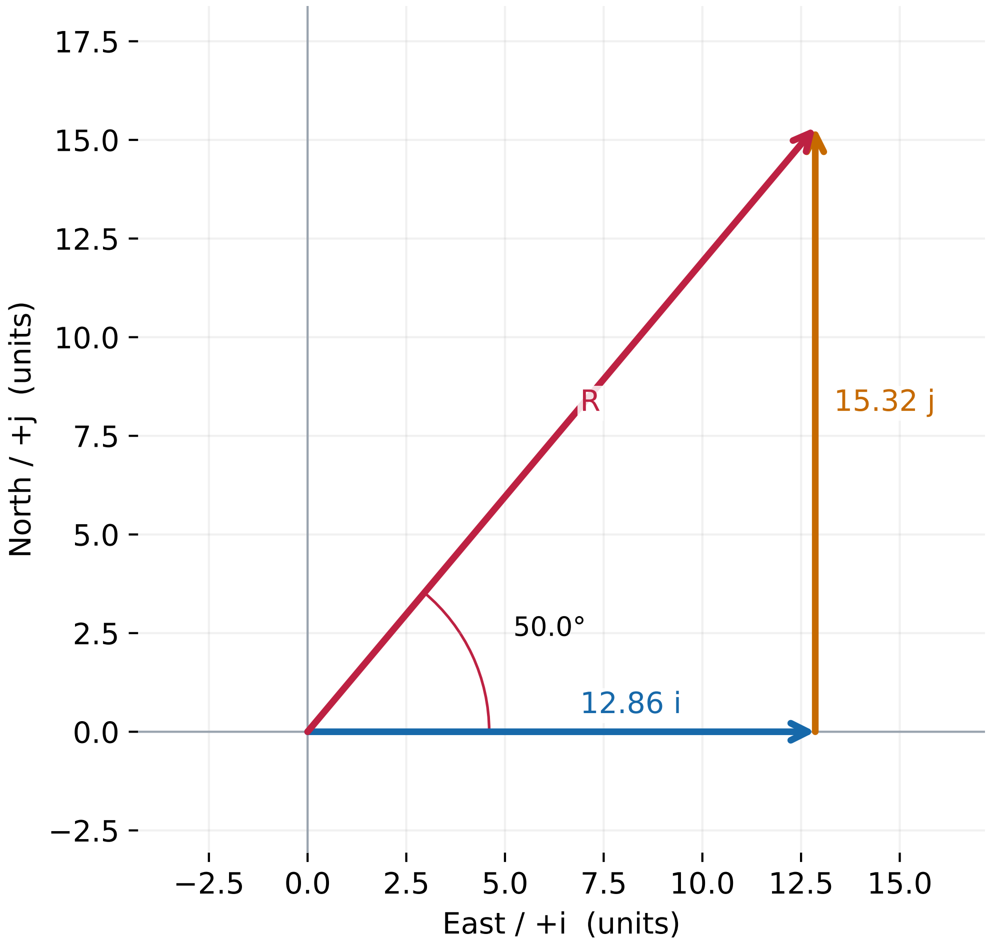
Equal horizontal and vertical scales. North is up; East is right.

Magnitude and trigonometry — problem 1



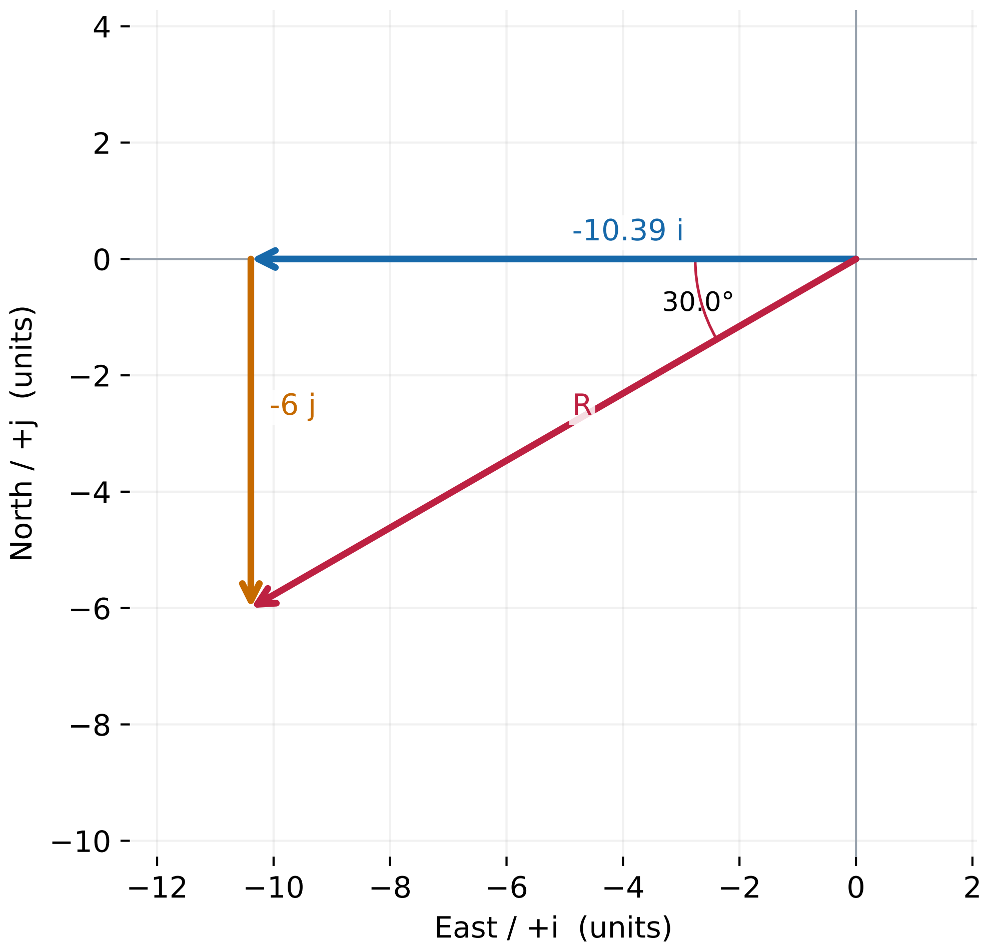
R = (4.00 i +5.00 j) units |R| = 6.40 units
51.3° N of E

Magnitude and trigonometry — problem 2



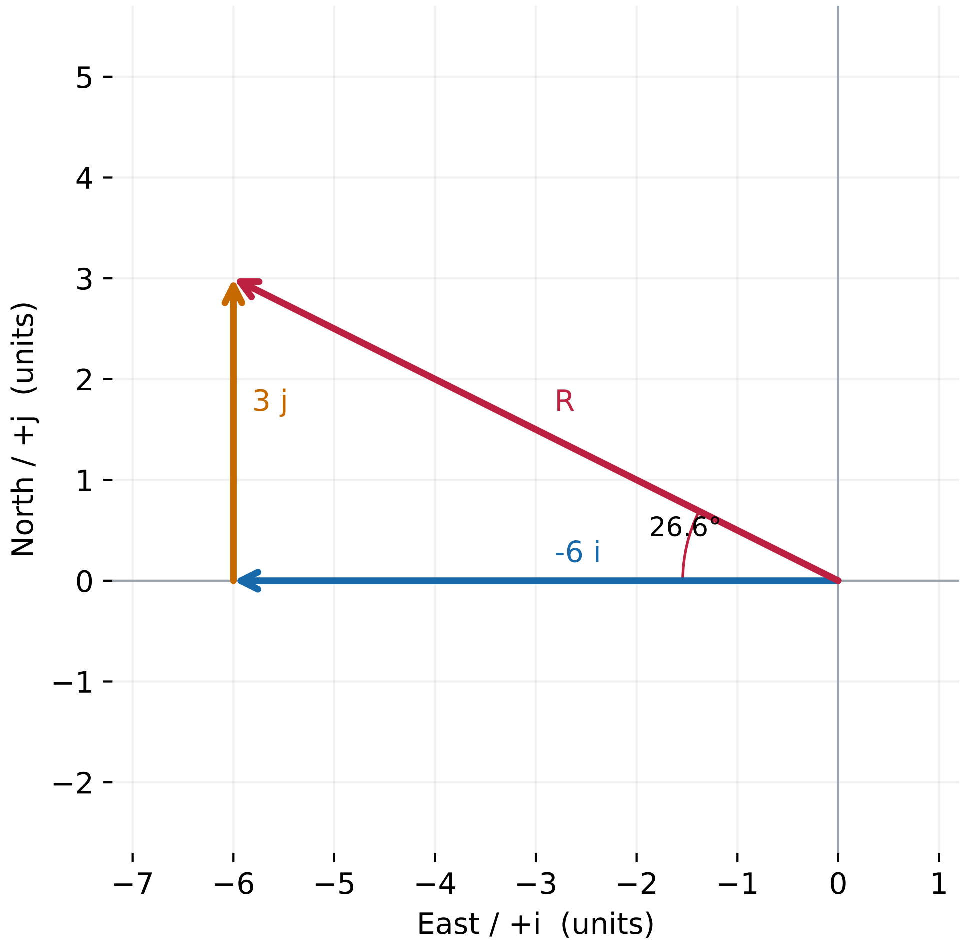
R = (12.86 i +15.32 j) units |R| = 20.00 units
50.0° N of E
Given 40° from vertical = 50° from horizontal.

Magnitude and trigonometry — problem 3



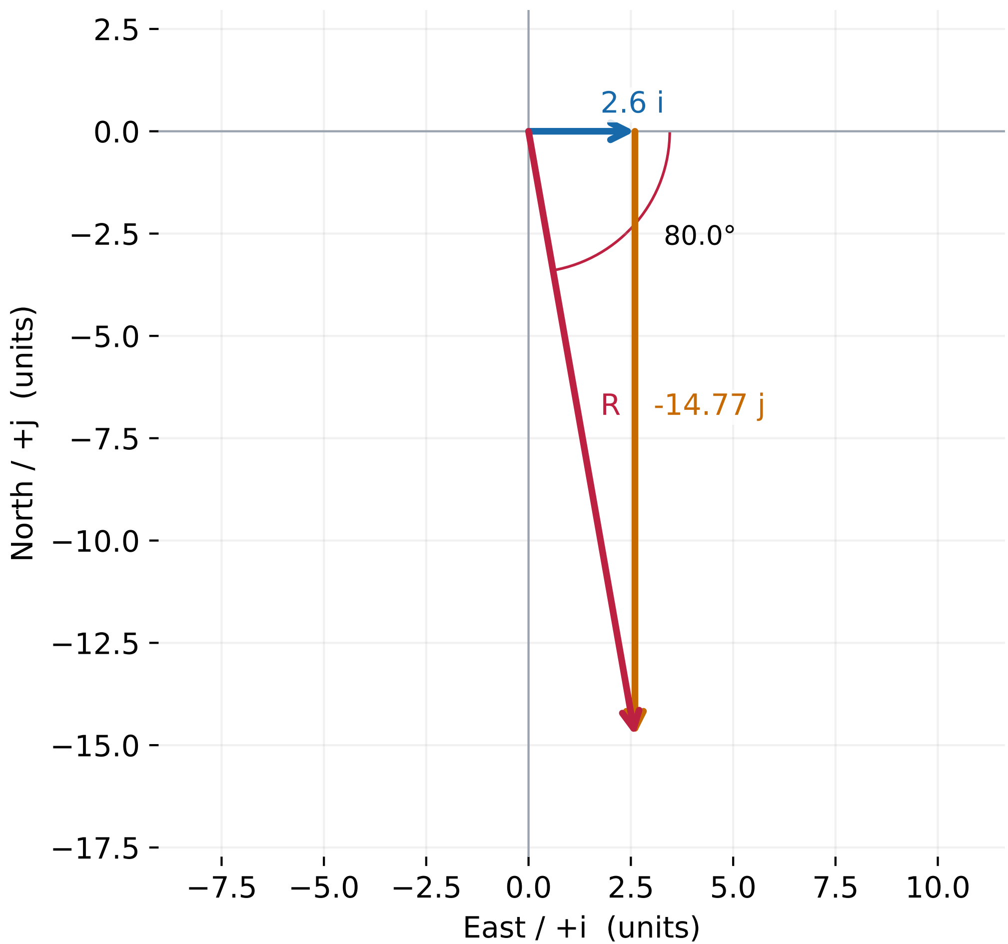
R = (-10.39 i -6.00 j) units |R| = 12.00 units
30.0° S of W

Magnitude and trigonometry — problem 4



R = (-6.00 i +3.00 j) units |R| = 6.71 units
26.6° N of W

Magnitude and trigonometry — problem 5

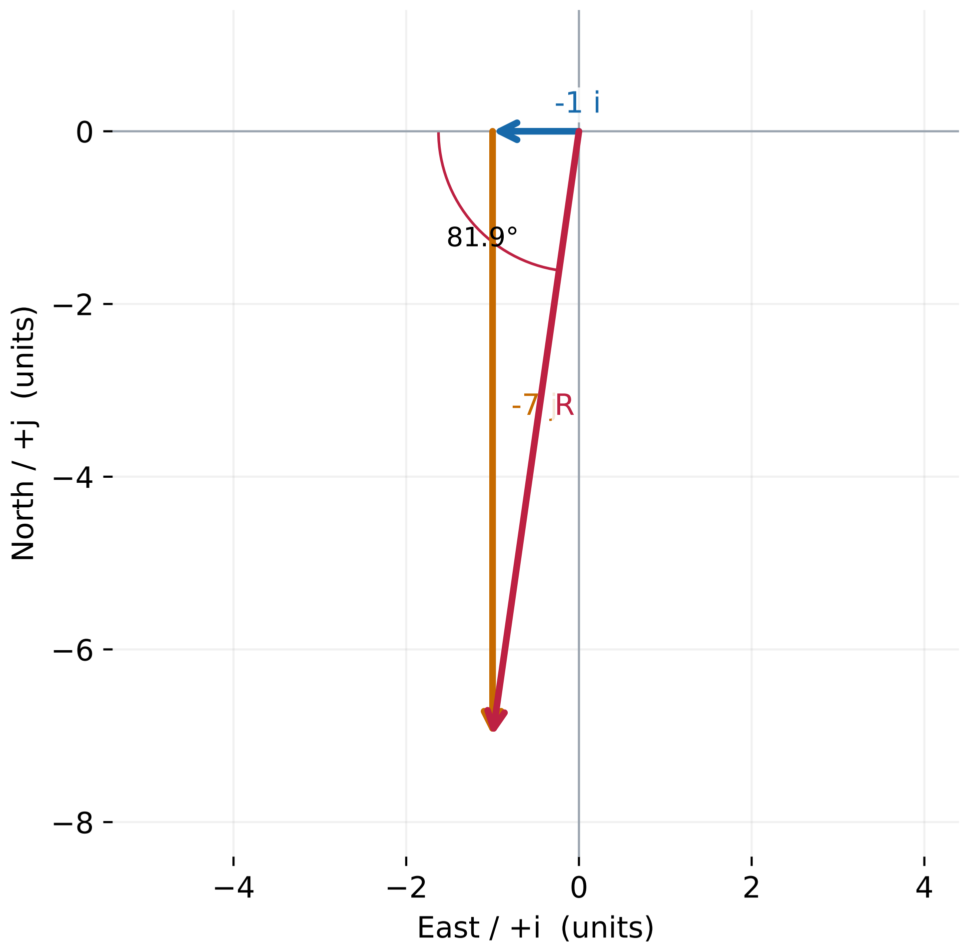


R = (2.60 i -14.77 j) units |R| = 15.00 units

80.0° S of E

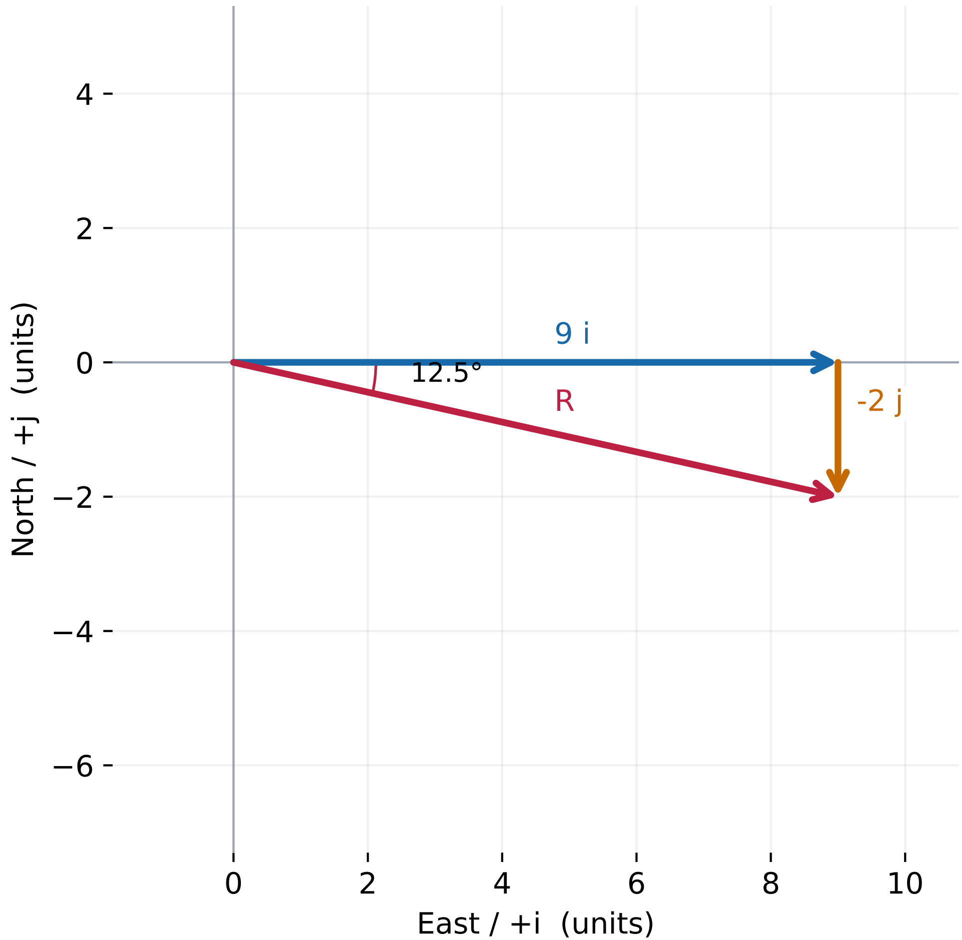
Source arrow points down and right: the x component is positive.

Magnitude and trigonometry — problem 6



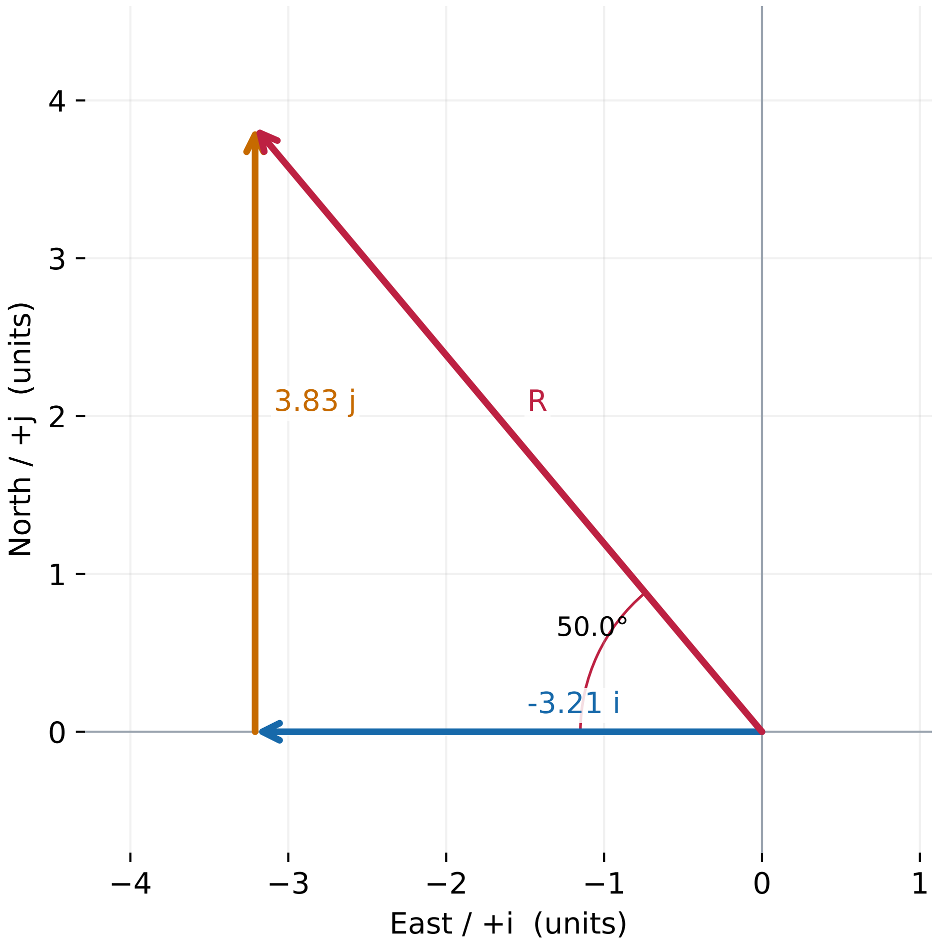
R = (-1.00 i -7.00 j) units |R| = 7.07 units
81.9° S of W

Magnitude and trigonometry — problem 7



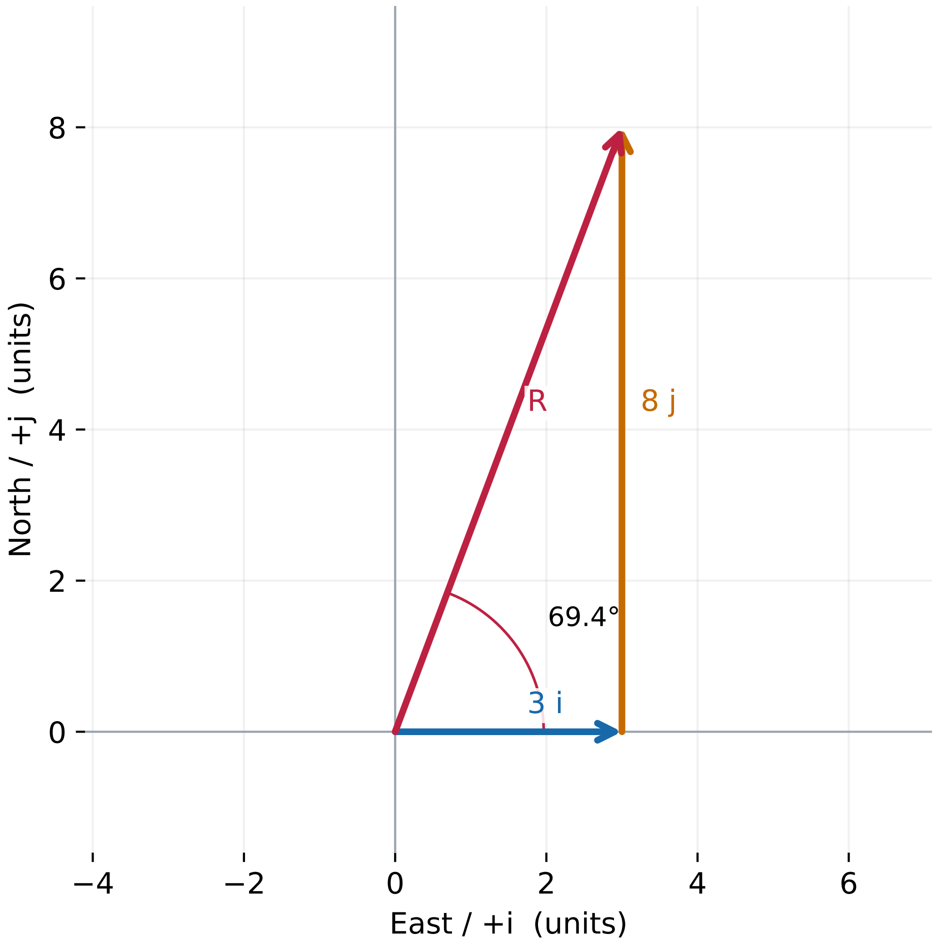
R = (9.00 i -2.00 j) units |R| = 9.22 units
12.5° S of E

Magnitude and trigonometry — problem 8



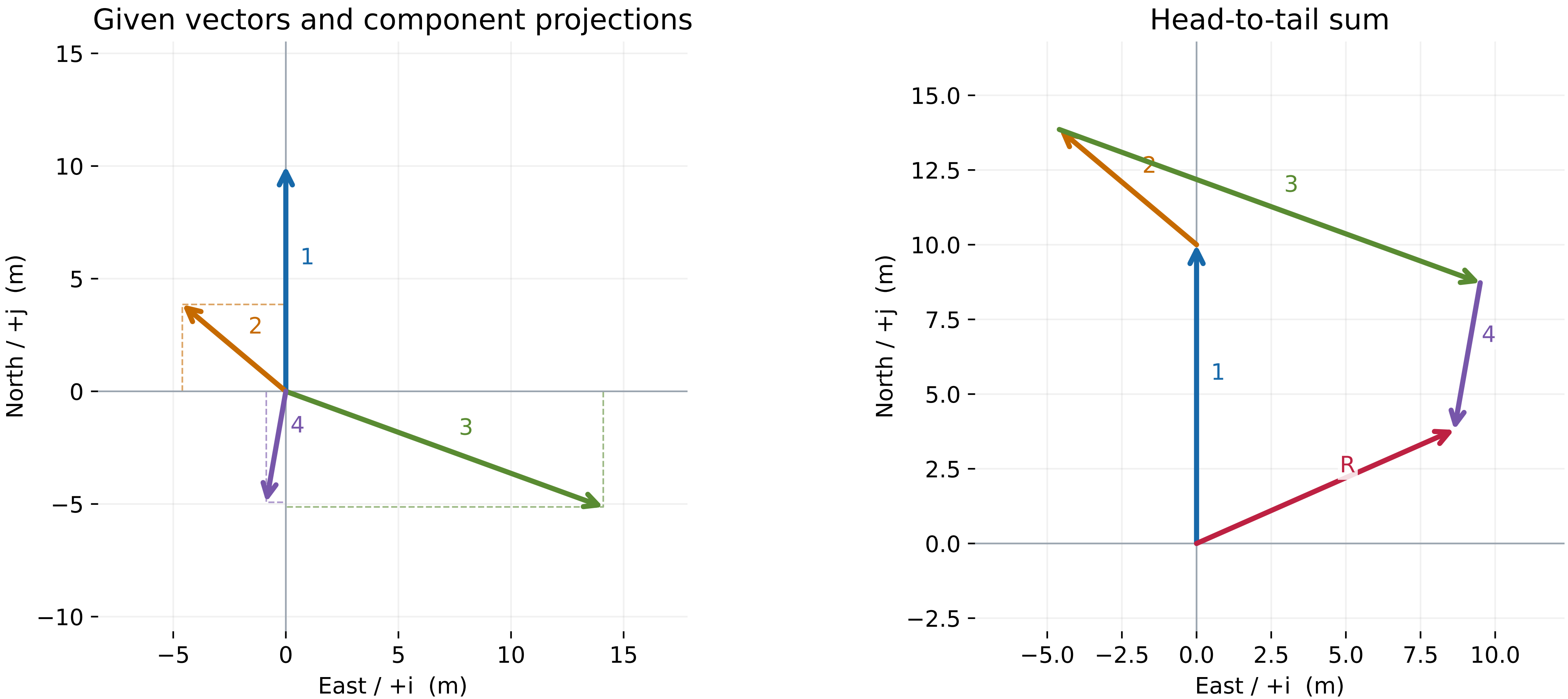
R = (-3.21 i +3.83 j) units |R| = 5.00 units
50.0° N of W

Magnitude and trigonometry — problem 9



R = (3.00 i +8.00 j) units |R| = 8.54 units
69.4° N of E

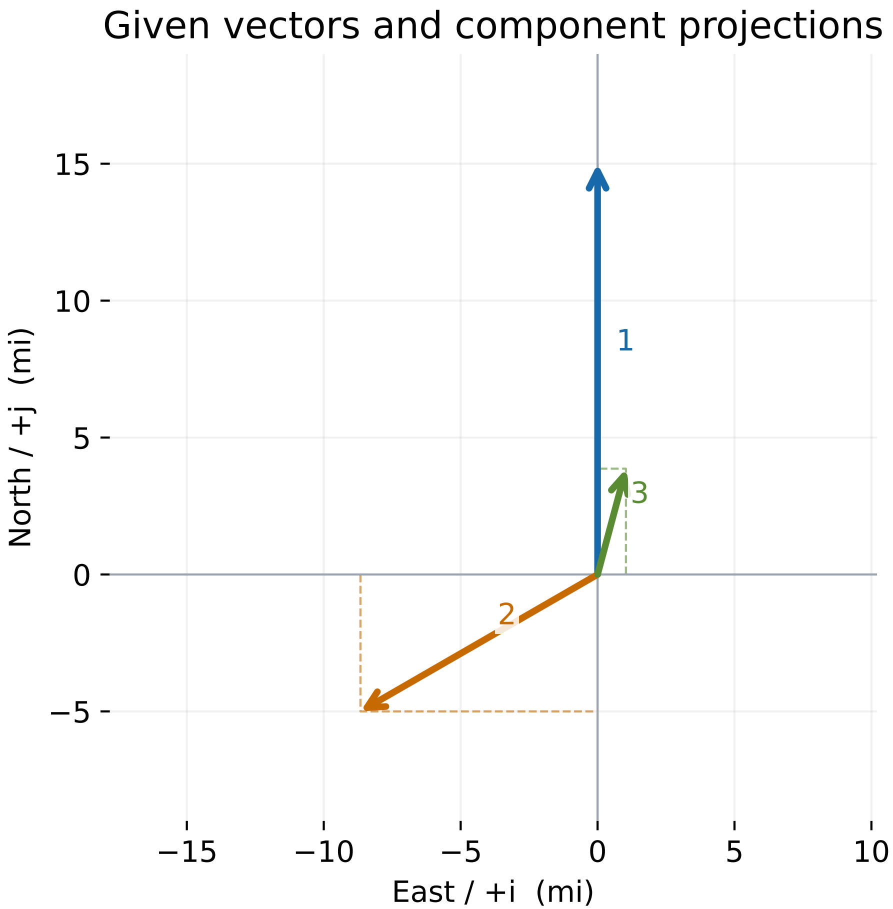
Multi-vector sum — problem 10



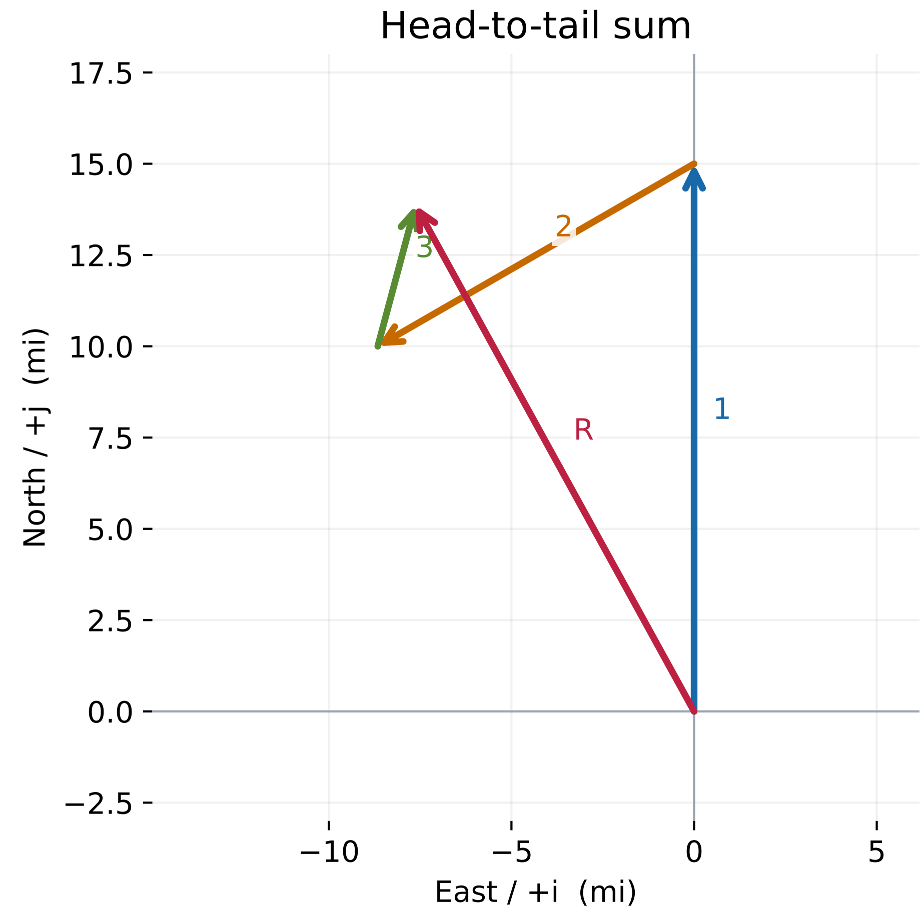
1. 10 m N: (0.00 i +10.00 j) m
2. 6 m, 40° N of W: (-4.60 i +3.86 j) m
3. 15 m, 20° S of E: (14.10 i -5.13 j) m
4. 5 m, 80° S of W: (-0.87 i -4.92 j) m

R = (8.63 i +3.80 j) m |R| = 9.43 m
23.8° N of E

Multi-vector sum — problem 11

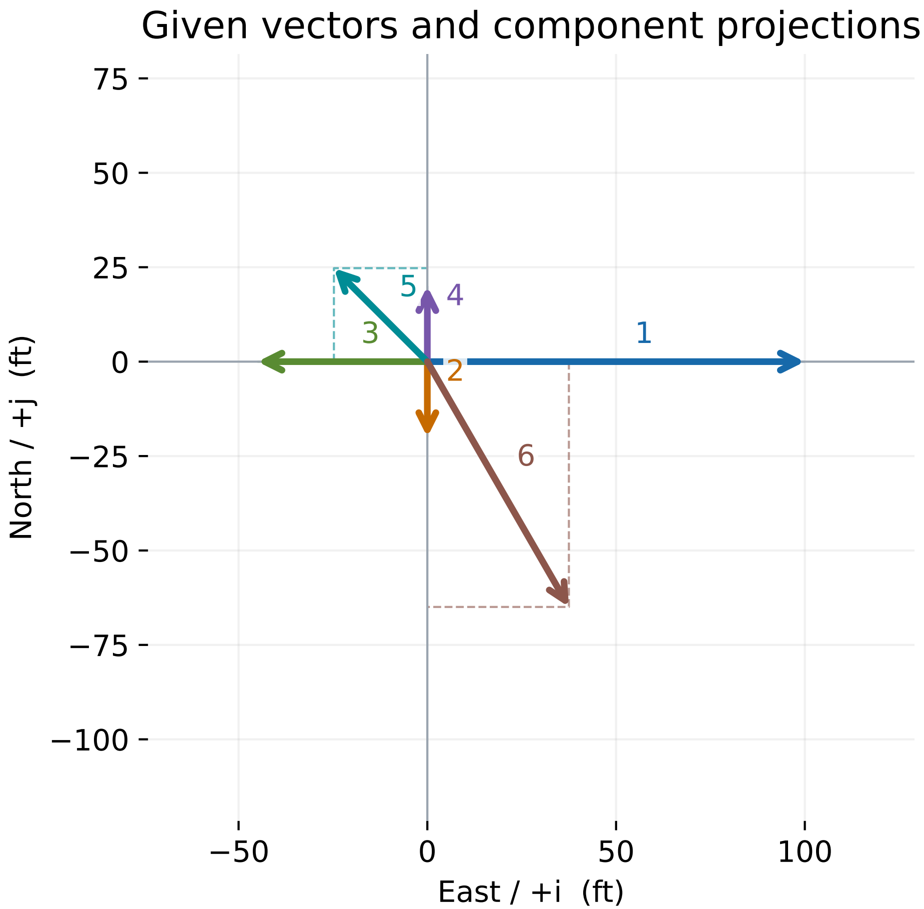


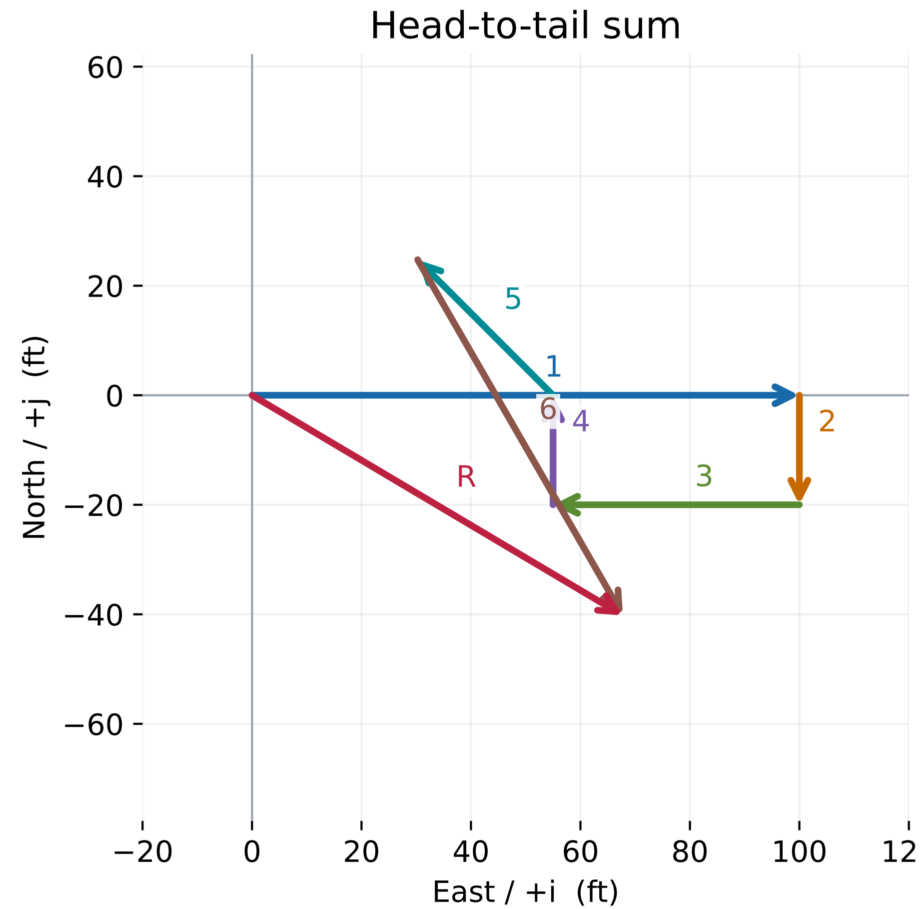
1. 15 mi N: (0.00 i +15.00 j) mi
2. 10 mi, 30° S of W: (-8.66 i -5.00 j) mi
3. 4 mi, 75° N of E: (1.04 i +3.86 j) mi



R = (-7.62 i +13.86 j) mi |R| = 15.82 mi
61.2° N of W

Multi-vector sum — problem 12





1. 100 ft E: (100.00 i +0.00 j) ft
2. 20 ft S: (0.00 i -20.00 j) ft
3. 45 ft W: (-45.00 i +0.00 j) ft
4. 20 ft N: (0.00 i +20.00 j) ft
5. 35 ft, 45° N of W: (-24.75 i +24.75 j) ft
6. 75 ft, 60° S of E: (37.50 i -64.95 j) ft

R = (67.75 i -40.20 j) ft |R| = 78.78 ft
30.7° S of E